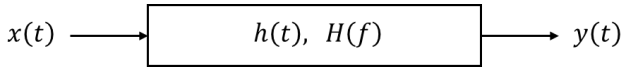
Homework: Time-frequency transform (2) – Obtain step response using Kaiser window in IFT

The S_{21} of a channel is typically a low-pass response due to dielectric and conductor losses; the attenuation at high frequencies is a natural "window" which helps mitigate the ripples in the time domain (TD) due to spectrum truncation. For S_{11}, however, it is usually a high-pass, because the effects of parasitic inductors and capacitors are more pronounced at high frequencies, causing more reflection. As such, computing reflected wave from limited spectrum is a more challenging task, and often requires explicit use of advanced windows like Kaiser.

This homework has two parts: in the first part, you are going to obtain the relation between the 10~90% risetime of the "integrated TD waveform of Kaiser window" with respect to the parameter α; this provides a foundation for choosing the parameter α in practical applications. In the second part, you are given the S parameters of a Chebyshev LPF (in Touchstone® SNP format); your task is to compute the step response of the LPF from its S_{11}, and compare the results with Ngspice simulation.

Theoretical note



For an LTI system, the output response is the convolution of the input with the impulse response:

$$y(t) = x(t) * h(t) = \int_{-\infty}^{\infty} h(\tau)x(t-\tau)d\tau$$

where the impulse response and the transfer function are Fourier transform pairs:

$$h(t) = \mathcal{F}^{-1}\{H(f)\} = \int_{-\infty}^{\infty} H(f)e^{j2\pi ft}df$$

When the input signal is the unit step function $u(t)$ (with risetime 0 and amplitude 1), the output is called the step response $s(t)$, which is useful in evaluating the system (such as TDR). As it turns out, the step response is the integral of the impulse response:

$$s(t) = h(t) * u(t) = \int_{-\infty}^{\infty} h(\tau)u(t-\tau)d\tau = \int_{-\infty}^{t} h(\tau)u(t-\tau)d\tau = \int_{-\infty}^{t} h(\tau)d\tau$$

In practice, we usually don't have the impulse response; instead, we only have the transfer function at discrete frequencies. What we could compute is the following:

$$h_W(t) \equiv \mathcal{F}^{-1}\{H(f)W(f)\} = \int_{-\infty}^{\infty} W(f)H(f)e^{j2\pi ft}df = \int_{-F/2}^{F/2} W(f)H(f)e^{j2\pi ft}df$$

where $W(f)$ is a window function which is identically zero outside the available bandwidth [−F/2, F/2]. From the convolution theorem, we can write

$$h_W(t) = h(t) * w(t)$$

where $w(t)$ is the TD waveform of the window. Therefore, the resulting $h_W(t)$ can be viewed as the *exact* output response of the system when the input signal is the TD waveform of the window $w(t)$. For

example, the frequency and time domain waveforms of the Kaiser window are

$$W(f) = \frac{I_0\left(\pi\alpha\sqrt{1-(2f/F)^2}\right)}{I_0(\pi\alpha)}$$

$$w(t) = \frac{F}{I_0(\pi\alpha)}\,\mathrm{sinc}\left(\sqrt{(Ft)^2-(\pi\alpha)^2}\right)$$

Note that the above $W(f)$ is centered at DC, which is different from the Kaiser window used for time-to-frequency transform that we used in the previous homework. Also note that $W(0) = 1$ in the above form of Kaiser window, which implies $\int_{-\infty}^{\infty} w(\tau)d\tau = 1$. Therefore, $w(t)$ is an approximated delta function, and $u_W(t) \equiv \int_{-\infty}^{t} w(\tau)d\tau$ is an approximated unit step with finite risetime. Now, if we integrate the $h_W(t)$, we get

$$\int_{-\infty}^{t} h_W(\tau)d\tau = \int_{-\infty}^{t} h(\tau) * w(\tau)\,d\tau = \int_{-\infty}^{t}\left[\int_{-\infty}^{\infty} h(\tau')w(\tau-\tau')d\tau'\right]d\tau$$

$$= \int_{-\infty}^{\infty} h(\tau')\left[\int_{-\infty}^{t} w(\tau-\tau')d\tau\right]d\tau' = \int_{-\infty}^{\infty} h(\tau')u_W(t-\tau')d\tau' = h(t) * u_W(t) =: s_W(t)$$

Hence, integrating $h_W(t)$ gives the output response when the input is $u_W(t)$, and is therefore an "approximate step response $s_W(t)$" (when the input step has finite risetime).

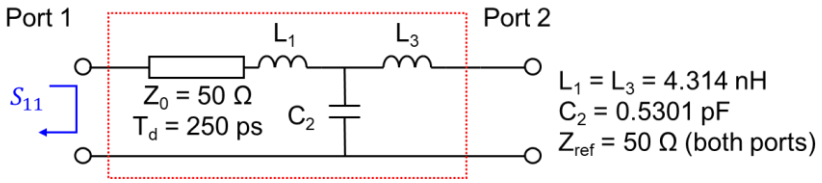
Part I:

(1) Plot the waveforms of $u_W(t)$ for $\alpha = 0, 1, 2, 3, 4$, on the same figure. The horizontal axis should be normalized by F, i.e., using $F{\cdot}t$ as the horizontal axis.

(2) Plot the normalized 10~90% risetime of $u_W(t)$ with respect to α.

Part II:

Consider the following circuit

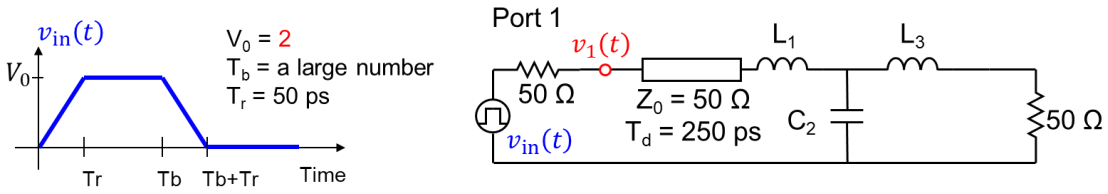


The L_1, C_2, L_3 together forms a 3rd-order 2-dB-ripple Chebyshev LPF with cutoff frequency 5 GHz and reference impedance 50 Ω at both ports. An ideal transmission line is added to port 1 to introduce a 250 ps delay in the reflected wave. The S_{11} of the overall circuit is given to you in the file `HW2_Chebyshev_LPF.s2p`. A Python module `snpreader.py` is also given to you which contains a function that could read general Touchstone® .snp files. Your task is to compute the step response of the reflected wave when the input step has a 10~90% risetime of 40 ps.

(1) Read the S parameters into Python, and plot dB(S_{11}) and dB(S_{21}) versus frequency on the same figure. (Note: as mentioned at the beginning, the S_{11} is a high-pass response, which is more difficult to compute time-domain responses given only limit spectrum.)

(2) Multiply the S_{11} by the Kaiser window, compute the $h_W(t)$, and then integrate $h_W(t)$ to obtain $s_W(t)$. Remember to choose an α such that the input step has a 10~90% risetime of 40 ps.

(3) Use Ngspice (or any other circuit simulator) to simulate the LPF in the time domain. Specifically, you should connect the circuits in the following way:



The input $v_{in}(t)$ is a pulse source in SPICE. The pulse amplitude should be 2 V because after voltage division the incident wave to the circuit will be 1 V. Also, the risetime of the pulse should be 50 ps such that the corresponding 10~90% risetime will be exactly 40 ps. Simulate the circuit to obtain the waveform of $v_1(t)$, which is the superposition of the incident wave $v_{in}(t)/2$ and the reflected wave $s_{Ng}(t)$. Process the result to extract $s_{Ng}(t)$, and plot it with $s_W(t)$ on the same figure. The results should be similar, with minor difference.

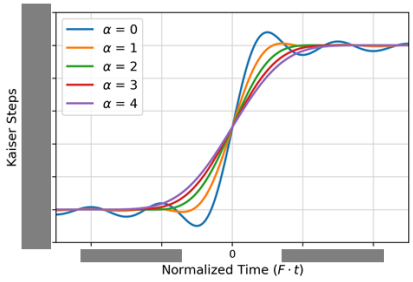
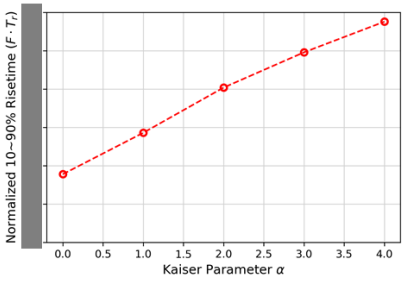
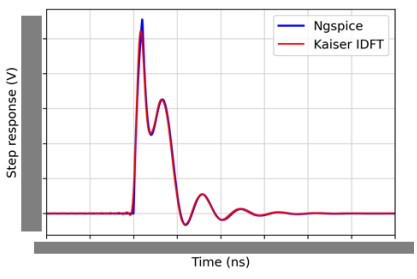
(4) Try different values of α, for example 0 and 4, and plot the resulting $s_W(t)$ against $s_{Ng}(t)$. You could see the effect of Kaiser window on reducing the ripples due to truncation, and effect of α on risetime.

Notes:

(1) The above Kaiser input waveform crosses the 50% point at $t = 0$, while in SPICE the pulse source crosses 50% at $t = T_r/2$. To compare the two results, you may add an additional delay T_1 to Kaiser. One simple way is to multiply the S_{11} by $\exp(-j\omega T_1)$ before IDFT. Due to the gradual rising of Kaiser from 0~10%, the exact value of T_1 is different from $T_r/2$. You can tune T_1 to obtain the best fit of $s_{Ng}(t)$.

(2) A Python sample code and Ngspice sample netlist are provided to you. You don't necessarily have to use them. However, you need to provide your source code in the report.

Appendix: reference answer

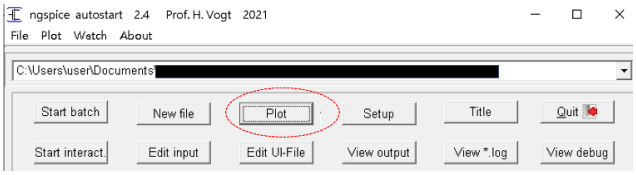
Part I (1)	Part I (2)	Part II (3)
		

It's just a reference. Your answers need not be the same.

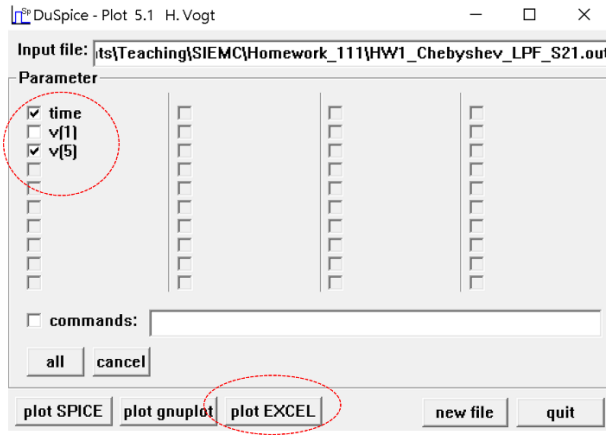
Appendix: How to export Ngspice's result to Python – simplest way

Step 1: Run simulation

Step 2: Plot



Step 3: Select variable and plot Excel



Step 4: go to the 3rd sheet



Step 5: Select the data columns needed.

8	Variables:		
9		0	1
10			
11		time	voltage
12		time	v(5)
13	0	0.00E+00	0.00E+00
14	1	1.00E-12	1.97E-16
15	2	2.00E-12	7.60E-15
16	3	3.00E-12	5.49E-14
17	4	4.00E-12	2.36E-13
18	5	5.00E-12	7.65E-13

Step 6: Copy the data to a new Excel file and save as csv

J20

	A	B	C
1	0.00E+00	0.00E+00	
2	1.00E-12	1.97E-16	
3	2.00E-12	7.60E-15	
4	3.00E-12	5.49E-14	
5	4.00E-12	2.36E-13	
6	5.00E-12	7.65E-13	
7	6.00E-12	2.05E-12	
8	7.00E-12	4.78E-12	

Step 7: read .csv into Python using

```
np.loadtxt('filename.csv',delimiter=',')
```